

Few Remarks on Series Tests

1. The Language of Asymptotic Functions

(i) We write $f(n) \sim g(n)$ and say $f(n)$ is **asymptotic** to $g(n)$ if

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1.$$

(ii) We write $f(n) \ll g(n)$ if

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0.$$

We say

- $f(n)$ is **negligible** compared to $g(n)$; or
- $g(n)$ **grows** faster than $f(n)$.

(iii) The list of basic types of functions (for the moment we call them **standard forms**):

- Constants such as $2, 3000, \pi, \sqrt{e}$.
- Constant positive powers of $\ln n$ such as $\ln n, \sqrt{\ln n}$.
- Constant positive powers of n such as $\sqrt{n}, n^2, n^{3/2}, n^e$.
- Exponentials c^n where $c > 1$ such as $(\frac{3}{2})^n, \pi^n$.
- Factorial $n!$.
- n^{cn} where $c > 0$ such as $n^n, n^{\frac{n}{4}}$.

(iv) There is a natural ordering of the standard forms:

$$\text{Constants} \ll (\ln n)^{c_1} \ll n^{c_2} \ll c_3^n \ll n! \ll n^{c_4 n}$$

where $c_1, c_2, c_4 > 0$ and $c_3 > 1$. Here we call it the **hierarchy**.

(v) We can also compare the functions of the same type by looking at their constants c_i . The one with larger c_i is bigger. Example: $(\ln n)^{1/2} \ll \ln n \ll (\ln n)^2$.

(vi) Some rules used on asymptotic functions.

Rule 1. If

- $f(n) = f_1(n) + f_2(n)$
- $f_1(n) \sim g(n)$
- $f_2(n) \ll g(n)$

then $f(n) \sim g(n)$.

Example:

- $n + \sqrt{n} \sim n$ (Ex. 9.4 Q18)
- $5n^3 - 3n \sim 5n^3$ (Ex. 9.4 Q24)
- $1 + \ln n \sim \ln n$ (Ex. 9.4 Q31)
- $\sqrt{e^x - x} \sim \sqrt{e^x}$ (Ex. 8.8 Q65)

Rule 2. If

- $f_1(n) \sim f_2(n)$
- $g_1(n) \sim g_2(n)$

then

- $f_1(n)g_1(n) \sim f_2(n)g_2(n)$
- $\frac{f_1(n)}{g_1(n)} \sim \frac{f_2(n)}{g_2(n)}$.

Example:

- $\frac{\sqrt{x+1}}{x^2} \sim \frac{1}{x^{3/2}}$ (Ex. 8.8 Q57)
- $\frac{2^n + 3^n}{3^n + 4^n} \sim \frac{3^n}{4^n}$ (Ex. 9.4 Q40)

2. How to use it in Series Tests?

- (i) Given a sequence $a_n = f(n)$, we want to compare $f(n)$ as a complicated function with a function $g(n)$ in standard form.
- (ii) If there is only one type of standard form in $f(n)$, obtain the function $g(n)$ and use Limit Comparison Test.

Example. Ex. 9.4 Q17. $\sum_{n=1}^{\infty} \frac{1}{2\sqrt{n} + \sqrt[3]{n}}$

Solution. Let $a_n = \frac{1}{2\sqrt{n} + \sqrt[3]{n}}$ and $b_n = \frac{1}{\sqrt{n}}$. Then $a_n > 0$ and $b_n > 0$ for all $n \geq 1$. Note that

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{2\sqrt{n} + \sqrt[3]{n}} = \lim_{n \rightarrow \infty} \frac{1}{2 + n^{-1/6}} = \frac{1}{2} > 0.$$

Since $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ is divergent (it is a p -series with $p = \frac{1}{2} \leq 1$), by the Limit Comparison Test, $\sum_{n=1}^{\infty} \frac{1}{2\sqrt{n} + \sqrt[3]{n}}$ diverges.

- (iii) If there are at least two types of standard forms appear as a product or a composition in $f(n)$, obtain the function $g(n)$ using the hierarchy and try to use Direct Comparison Test.

Example. Ex. 9.4 Q27. $\sum_{n=3}^{\infty} \frac{1}{\ln(\ln n)}$.

Solution. Since $\ln x \leq x$ for $x \geq 1$, we have

$$0 \leq \ln(\ln n) \leq \ln n \leq n \quad \text{for all } n \geq 3,$$

we get

$$\frac{1}{\ln(\ln n)} \geq \frac{1}{n} \geq 0 \quad \text{for all } n \geq 3.$$

Since $\sum_{n=3}^{\infty} \frac{1}{n}$ is divergent, by the Direct Comparison Test, $\sum_{n=3}^{\infty} \frac{1}{\ln(\ln n)}$ diverges.

Remark 1. If there is a sum of two standard forms $f(n) + g(n)$ with $f(n) \ll g(n)$, then $f(n) + g(n) \sim g(n)$ by using Rule 1.

Remark 2. Sometimes we may try Ratio Test. It depends on the context (explained in the next slide).

3. When to use Ratio Test or Root Test?

If $a_n = f(n)$ only contains sums, products or quotients of standard forms, then we usually use Ratio Test. Otherwise, use Root Test.

Examples.

Ex. 9.5 Q6. $\sum_{n=2}^{\infty} \frac{3^{n+2}}{\ln n} \Rightarrow$ Ratio Test

Ex. 9.5 Q7. $\sum_{n=1}^{\infty} (-1)^n \frac{n^2(n+2)!}{n!3^{2n}} \Rightarrow$ Ratio Test

Ex. 9.5 Q9. $\sum_{n=1}^{\infty} \frac{7}{(2n+5)^n} \Rightarrow$ Root Test

Ex. 9.5 Q14. $\sum_{n=1}^{\infty} \sin^n \left(\frac{1}{\sqrt{n}} \right) \Rightarrow$ Root Test

Ex. 9.5 Q31. $\sum_{n=1}^{\infty} \frac{e^n}{n^e} \Rightarrow$ Ratio Test

Remark. Ratio/Root Test does not work if a_n only contains powers of n or logarithms.